

Physics-Informed, GAN-Augmented Graph Neural Networks for Multi-Horizon Dengue Forecasting

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Abstract

Dengue is the fastest-spreading mosquito-borne viral disease in the world, and Sri Lanka carries a heavy, growing burden. Accurate multi-week, district-level forecasts would let health authorities pre-position resources and act before outbreaks peak, but the task is hard: transmission is coupled across districts through human movement, the data are scarce (25 districts over roughly a decade of weekly counts), and purely data-driven models can produce epidemiologically implausible forecasts. We build a spatio-temporal Graph Neural Network (GNN) baseline for weekly dengue incidence and propose three stackable contributions: an epidemic-mechanistic (SEIR-SEI) *physics-informed loss*, *GAN-based augmentation* to counter scarcity, and a *learned adaptive graph*. To our knowledge this combination is unexplored for dengue. Under rolling-origin cross-validation, our GCN/GAT baseline matches a strong persistence floor (RMSE ≈ 45) - the honest starting point our contributions aim to improve.

Keywords

dengue forecasting, graph neural networks, physics-informed learning, data augmentation, spatio-temporal modeling

1 Problem Motivation

Dengue remains one of the most important public-health problems in tropical regions, and Sri Lanka carries a persistent, growing burden: the 2017 epidemic alone produced over 186,000 reported cases and 440 deaths, and case counts have trended upward since the 2000s despite sustained vector control. Because the interval between infection and diagnosis is on the order of three weeks, accurate multi-week-ahead forecasting of district-level incidence is directly actionable - it gives the Ministry of Health and regional teams a window to pre-position resources, intensify larval-source reduction, and issue early warnings before an outbreak peaks. The core problem we address is **high-accuracy, multi-horizon spatio-temporal forecasting** under conditions that make it genuinely hard.

Three intertwined difficulties motivate our work. First, transmission is a coupled **spatio-temporal** process: incidence in one district depends on its own climatic and epidemiological history and on transmission spilling in from neighbouring districts through human movement. Classical statistical models (ARIMA/SARIMA) and non-spatial machine learning treat districts independently and cannot represent this structure. GNNs close much of this gap by modelling districts as connected nodes, and prior work confirms they outperform traditional early-warning systems on this exact Sri Lanka dataset [1], but that work applies off-the-shelf graph models and explicitly leaves methodological improvement open. Second,

the problem is severely **data-limited**: 25 districts at weekly resolution over roughly a decade, a small graph and short history relative to what deep models need, which encourages overfitting. Third, purely data-driven graph models can produce **epidemiologically implausible** forecasts - hallucinated spikes propagated across the graph - with no guarantee that predictions respect disease-spread dynamics, undermining operational trust.

These gaps map directly onto our contributions: an improved graph architecture for accuracy, a physics-informed loss constraining forecasts toward host-vector dynamics, and GAN augmentation to counter scarcity. The **beneficiaries** are Sri Lanka's National Dengue Control Unit and district health teams - who gain earlier, more reliable, more interpretable signals - and the communities whose exposure to severe dengue can be reduced through timely response.

2 Dataset

Primary - Sri Lanka district-level dengue. We use a multi-source collection aligning epidemiological, meteorological, and geospatial data across Sri Lanka's **25 districts** at **weekly** resolution over **2013-2022** [1]. Weekly case counts come from the Ministry of Health / Epidemiology Unit; environmental covariates (precipitation, air and land-surface temperature, humidity, NDVI, soil moisture, canopy) are drawn from NASA EarthData satellite products aggregated to weekly resolution. District boundaries come from the GADM shapefile, each district a GIS polygon. The processed array has shape (459 weeks, 25 districts, 11 features). Environmental drivers are lag-shifted by their empirically optimal delay (e.g. precipitation ≈ 12 weeks, minimum NDVI ≈ 17 weeks), reflecting mosquito-breeding and infection-to-diagnosis intervals.

Graph structure. Districts are nodes; edges encode inter-district adjacency. Because reliable mobility data are unavailable, the reference adjacency is built from district-to-district distances (with self-loops, 141 directed edges). Replacing this with a *learned adaptive graph* is one of our contributions.

Scale and scarcity. The target is extremely heavy-tailed (median 13 cases, max 2631, 9.7% zeros) with strong week-to-week autocorrelation (lag-1 ≈ 0.68). Relative to benchmark graphs (e.g. METR-LA, 207 nodes), the small node count and short history make overfitting a real risk - motivating our GAN augmentation and physics-informed regularization. As an optional generalization test we may additionally evaluate on OpenDengue [2], a standardized repository of >56M dengue records across 102 countries.

3 Baseline Model

Architecture. The mandated graph baseline is a spatio-temporal GNN with a shared *spatial encoder* $\rightarrow$ *prediction head* design. Two

graph-convolution layers (GCN [3], or GAT [4] with 8 attention heads) aggregate information across neighbouring districts; a ReLU + dropout + linear head maps each node embedding to $H=3$ future weekly values. Per district, the last $W=3$ weeks of all 11 features form a 33-dimensional node input.

Implementation. Implemented in PyTorch Geometric; we forecast 1-, 2-, and 3-weeks ahead. Two training refinements were essential to obtain a *competitive* baseline on this heavy-tailed, highly auto-correlated data: (i) modelling cases in **log1p** space, and (ii) **residual-over-persistence** learning - the network predicts the correction to last week’s value rather than the absolute count - with gradient clipping. Features are z-normalized using *training-weeks statistics only* (leakage-free). Optimization uses Adam ($\text{lr}=10^{-3}$, weight decay 5×10^{-4}) with early stopping on validation RMSE.

Evaluation protocol. A single 70/10/20 chronological split proved unreliable: validation and test RMSE were anti-correlated ($r \approx -0.74$) because the two periods fall in different epidemic regimes, so selection on one slice mis-selected for the other. We therefore adopt rolling-origin (expanding-window) cross-validation [17] - retraining at three chronological origins and averaging over 3 seeds - the correct protocol for non-stationary series.

Hyperparameter tuning. Grid search over learning rate $\{10^{-2}, 10^{-3}, 5 \times 10^{-4}\}$, hidden size $\{32, 64\}$, and dropout $\{0, 0.1, 0.3\}$, selected on validation RMSE per fold; the best configuration was $\text{lr}=10^{-3}$, hidden 64.

Results. Table 1 reports rolling-origin test performance: both GCN and GAT match a strong persistence baseline overall (RMSE ≈ 45 vs 44.8), and on the most recent fold they surpass it (GCN 66.4 vs persistence 68.6). Table 2 shows residual-over-persistence is what lifts the plain GNN from RMSE ≈ 66 to ≈ 45 ; log-space adds a modest MAE gain. Qualitatively, the tuned model successfully tracks outbreak dynamics in high-incidence districts like Colombo, demonstrating that the spatio-temporal GNN **matches the persistence floor** - the honest starting point our contributions must improve.

Table 1: Baseline rolling-origin test performance (mean over 3 folds, 3 seeds).

Model	RMSE	MAE
Persistence (naive reference)	44.8	15.7
GCN (residual + log)	45.3	15.9
GAT (residual + log)	45.5	15.9

Table 2: Ablation of baseline training refinements (GCN, rolling-origin).

Setting	RMSE	MAE
Plain GCN (absolute scale)	66.2	31.6
+ log1p only	67.1	27.0
+ residual only	45.0	16.8
residual + log	45.3	15.8

4 Proposed Contribution

We push the baseline *below* the persistence floor via three stackable contributions, each attaching at a clean seam of the baseline.

(a) Physics-informed loss (novel training strategy). We add a soft-constraint term penalizing deviation from an **SEIR-SEI** host-vector compartmental update [12] - the standard mechanistic model for dengue - with population-conservation and neighbour-smoothness constraints across the district graph: $\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_p \mathcal{L}_{\text{phys}} + \lambda_c \mathcal{L}_{\text{cons}} + \lambda_s \mathcal{L}_{\text{smooth}}$. This follows the epidemiology-informed neural-network line [9–11] and is expected to improve robustness, peak timing, and extrapolation under distribution shift - exactly where a mean-reverting data-driven model fails.

(b) GAN-based augmentation (novel data strategy). To counter scarcity, we synthesize additional (window $\rightarrow$ horizon) training sequences with a conditional time-series GAN (TimeGAN [13]- or RCGAN [14]-style, conditioned on covariates and outbreak labels) and measure downstream gains, always against simpler augmentation (jittering, window-warping) so the GAN must earn its complexity.

(c) Adaptive graph architecture (novel representation). We replace the hand-built distance adjacency with a *learned adaptive* adjacency (Graph WaveNet [5]-style node embeddings), optionally with a gravity-mobility prior, so connectivity is inferred from data.

Expected novelty. Our contribution is an architecture that uses a mechanistic SEIR-SEI loss to regularize GAN-augmented STGNNs. By resolving the fundamental algorithmic conflict between unconstrained data-driven forecasting and rigid mechanistic limits, this framework advances the Sri Lanka Dengue benchmark well beyond the application of standard baselines [1].

Expected improvement. We target consistent RMSE/MAE reductions below the persistence floor and, importantly, sharper, better-calibrated probabilistic forecasts and improved peak-onset detection - the outputs that matter for early warning. Because our baseline already matches a strong naive method, any measured gain is a genuine improvement rather than the rescue of a weak baseline.

5 Experimental Plan

Evaluation metrics: Point accuracy: RMSE, MAE, and SMAPE (plus masked MAPE on non-zero weeks, since frequent zero-case weeks make plain MAPE unusable). Probabilistic quality: CRPS and prediction-interval coverage/width (PICP/MPIW) [16]. Spatial quality: Moran’s I on residuals, to confirm genuine spatial autocorrelation without spurious spillover. All metrics are reported per horizon.

Baselines: Naive persistence; statistical (ARIMA, SARIMA); machine learning (Random Forest, XGBoost); deep sequential (LSTM, GRU, ConvLSTM); and graph (DCRNN [6], STGCN, EpiGNN [7], ColaGNN [8], and our GCN/GAT). Contributions are compared against these and against each other via a stacked ablation (baseline $\rightarrow$ +physics $\rightarrow$ +GAN $\rightarrow$ +adaptive graph) under the identical rolling-origin protocol. **Resource requirements:** All experiments are designed for **free Google Colab / Kaggle GPU**. The graph is small (25 nodes, 141 edges) and each fold trains in minutes on a single GPU; the GAN is the only moderately heavier component and stays within free-tier budgets. No multi-GPU, HPC, or paid cloud resources are assumed.

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